

Employees Table:

```
18 );
19 • INSERT INTO Employees (EmployeeID, FirstName, LastName, DepartmentID)
20 VALUES
21 (1, 'John', 'Doe', 1),
22 (2, 'Jane', 'Smith', 2),
23 (3, 'Sam', 'Brown', NULL);
24 • INSERT INTO Departments (DepartmentID, DepartmentName) VALUES
25 (1, 'HR'),
26 (2, 'IT'),
27 (3, 'Sales');
28 • INSERT INTO Projects (ProjectID, ProjectName, EmployeeID) VALUES
29 (101, 'Project A', 1),
30 (102, 'Project B', 2),
31 (103, 'Project C', 2),
32 (104, 'Project D', NULL);
33
34 • select * from Employees;
35
```

100% 25:34

Result Grid Filter Rows: Search Edit: Export/Import:

	EmployeeID	FirstName	LastName	DepartmentID
1	1	John	Doe	1
2	2	Jane	Smith	2
3	3	Sam	Brown	NULL
	NULL	NULL	NULL	NULL

Departments Table:

```
19 • INSERT INTO Employees (EmployeeID, FirstName, LastName, DepartmentID)
20 VALUES
21 (1, 'John', 'Doe', 1),
22 (2, 'Jane', 'Smith', 2),
23 (3, 'Sam', 'Brown', NULL);
24 • INSERT INTO Departments (DepartmentID, DepartmentName) VALUES
25 (1, 'HR'),
26 (2, 'IT'),
27 (3, 'Sales');
28 • INSERT INTO Projects (ProjectID, ProjectName, EmployeeID) VALUES
29 (101, 'Project A', 1),
30 (102, 'Project B', 2),
31 (103, 'Project C', 2),
32 (104, 'Project D', NULL);
33
34 • select * from Employees;
35 • select * from Departments;
36
```

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Result Grid Filter Rows: Search Edit: Export/Import:

	DepartmentID	DepartmentNa...	
	1	HR	
	2	IT	
	3	Sales	
	NULL	NULL	

Departments 8

Projects Table:

```
20 VALUES
21 (1, 'John', 'Doe', 1),
22 (2, 'Jane', 'Smith', 2),
23 (3, 'Sam', 'Brown', NULL);
24 INSERT INTO Departments (DepartmentID, DepartmentName) VALUES
25 (1, 'HR'),
26 (2, 'IT'),
27 (3, 'Sales');
28 INSERT INTO Projects (ProjectID, ProjectName, EmployeeID) VALUES
29 (101, 'Project A', 1),
30 (102, 'Project B', 2),
31 (103, 'Project C', 2),
32 (104, 'Project D', NULL);
33
34 select * from Employees;
35 select * from Departments;
36 select * from Projects;
37
```

100% 24:36

Result Grid Filter Rows: Search Edit: Export/Import:

ProjectID	ProjectName	EmployeeID
101	Project A	1
102	Project B	2
103	Project C	2
104	Project D	NULL
NULL	NULL	NULL

1. INNER JOIN

Write a query to retrieve all employees along with their department names. Only include employees who are assigned to a department.

```
select e.employeeid, e.firstname, e.lastname, d.departmentname
from employees e
inner join departments d on e.departmentid = d.departmentid;
```

```
25 (1, 'HR'),
26 (2, 'IT'),
27 (3, 'Sales');
28 INSERT INTO Projects (ProjectID, ProjectName, EmployeeID) VALUES
29 (101, 'Project A', 1),
30 (102, 'Project B', 2),
31 (103, 'Project C', 2),
32 (104, 'Project D', NULL);
33
34 select * from Employees;
35 select * from Departments;
36 select * from Projects;
37
38 select e.employeeid, e.firstname, e.lastname, d.departmentname
39 from employees e
40 inner join departments d on e.departmentid = d.departmentid;
```

100% 61:40

Result Grid Filter Rows: Search Export:

	employeeid	firstname	lastname	departmentna...
1	1	John	Doe	HR
2	2	Jane	Smith	IT

2.

Write a query to retrieve all employees along with their department names. Include employees even if they are not assigned to a department.

```
select e.employeeid, e.firstname, e.lastname, d.departmentname
from employees e
left join departments d on e.departmentid = d.departmentid;
```

```
32 (104, 'Project D', NULL);
33
34 • select * from Employees;
35 • select * from Departments;
36 • select * from Projects;
37
38 • select e.employeeid, e.firstname, e.lastname, d.departmentname
39   from employees e
40  inner join departments d on e.departmentid = d.departmentid;
41
42
43 • select e.employeeid, e.firstname, e.lastname, d.departmentname
44   from employees e
45  left join departments d on e.departmentid = d.departmentid;
46
47
48
49
```

100% 60:45

Result Grid Filter Rows: Search Export:

	employeeid	firstname	lastname	departmentna...
1	1	John	Doe	HR
2	2	Jane	Smith	IT
3	3	Sam	Brown	NULL

3.

Write a query to retrieve all departments along with the employees in each department. Include departments even if no employees are assigned to them.

```
select d.departmentid, d.departmentname, e.employeeid, e.firstname, e.lastname
from departments d
left join employees e on d.departmentid = e.departmentid;
```

```

32 (104, 'Project D', NULL);
33
34 • select * from Employees;
35 • select * from Departments;
36 • select * from Projects;
37
38 • select e.employeeid, e.firstname, e.lastname, d.departmentname
39   from employees e
40  inner join departments d on e.departmentid = d.departmentid;
41
42
43 • select e.employeeid, e.firstname, e.lastname, d.departmentname
44   from employees e
45  left join departments d on e.departmentid = d.departmentid;
46
47 • select d.departmentid, d.departmentname, e.employeeid, e.firstname, e.lastname
48   from departments d
49  left join employees e on d.departmentid = e.departmentid;
50
100% 58:49
Result Grid Filter Rows: Search Export:

```

	department...	departmentna...	employeeid	firstname	lastname
1		HR	1	John	Doe
2		IT	2	Jane	Smith
3		Sales	NULL	NULL	NULL

4.

Write a query to retrieve all employees and all departments. Include employees even if they are

not assigned to a department and departments even if they have no employees

```

select e.employeeid, e.firstname, e.lastname, d.departmentname
from employees e
left join departments d on e.departmentid = d.departmentid
union

```

```
select e.employeeid, e.firstname, e.lastname, d.departmentname
from employees e
right join departments d on e.departmentid = d.departmentid;
```

```
34 • select * from Employees;
35 • select * from Departments;
36 • select * from Projects;
37
38 • select e.employeeid, e.firstname, e.lastname, d.departmentname
39   from employees e
40   inner join departments d on e.departmentid = d.departmentid;
41
42
43 • select e.employeeid, e.firstname, e.lastname, d.departmentname
44   from employees e
45   left join departments d on e.departmentid = d.departmentid;
46
47 • select d.departmentid, d.departmentname, e.employeeid, e.firstname, e.lastname
48   from departments d
49   left join employees e on d.departmentid = e.departmentid;
50
51 • select e.employeeid, e.firstname, e.lastname, d.departmentname
52   from employees e
```

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Result Grid Filter Rows: Search Export:

	employeeid	firstname	lastname	departmentna...
1	1	John	Doe	HR
2	2	Jane	Smith	IT
3	3	Sam	Brown	NULL
	NULL	NULL	NULL	Sales

5.

Write a query to retrieve all possible combinations of employees and projects

```
select e.employeeid, e.firstname, e.lastname, p.projectid, p.projectname
from employees e
cross join projects p;
```

```
46
47 ■ select d.departmentid, d.departmentname, e.employeeid, e.firstname, e.lastname
48     from departments d
49     left join employees e on d.departmentid = e.departmentid;
50
51 ■ select e.employeeid, e.firstname, e.lastname, d.departmentname
52     from employees e
53     left join departments d on e.departmentid = d.departmentid
54     union
55     select e.employeeid, e.firstname, e.lastname, d.departmentname
56     from employees e
57     right join departments d on e.departmentid = d.departmentid;
58
59 ■ select e.employeeid, e.firstname, e.lastname, p.projectid, p.projectname
60     from employees e
61     cross join projects p;
62
63
```

100% 23:61

Result Grid Filter Rows: Search Export:

	employeeid	firstname	lastname	projectid	projectname
	3	Sam	Brown	101	Project A
	2	Jane	Smith	101	Project A
	1	John	Doe	101	Project A
	3	Sam	Brown	102	Project B
	2	Jane	Smith	102	Project B
	1	John	Doe	102	Project B
	3	Sam	Brown	103	Project C
	2	Jane	Smith	103	Project C
	1	John	Doe	103	Project C

Result 10

6.

Write a query to retrieve all employees, their department names, and the projects they are working on. Include employees even if they are not assigned to a department or project

```
select e.employeeid, e.firstname, e.lastname, d.departmentname, p.projectname
from employees e
left join departments d on e.departmentid = d.departmentid
left join projects p on e.employeeid = p.employeeid;
```



```
select d.departmentid, d.departmentname
from departments d
left join employees e on d.departmentid = e.departmentid
where e.employeeid is null;
```

[illegible]

```
select p.projectid, p.projectname
from projects p
left join employees e on p.employeeid = e.employeeid
where p.employeeid is null;
```

```

63 ■ select e.employeeid, e.firstname, e.lastname, d.departmentname, p.projectname
64     from employees e
65     left join departments d on e.departmentid = d.departmentid
66     left join projects p on e.employeeid = p.employeeid;
67
68
69 ■ select d.departmentid, d.departmentname
70     from departments d
71     left join employees e on d.departmentid = e.departmentid
72     where e.employeeid is null;
73
74 ■ select p.projectid, p.projectname
75     from projects p
76     left join employees e on p.employeeid = e.employeeid
77     where p.employeeid is null;
78
79
80

```

100% 28:77

Result Grid



Filter Rows:



Search

Export:



projectid	projectname	
104	Project D	

Result 14

Result 15